Coded Beam Training

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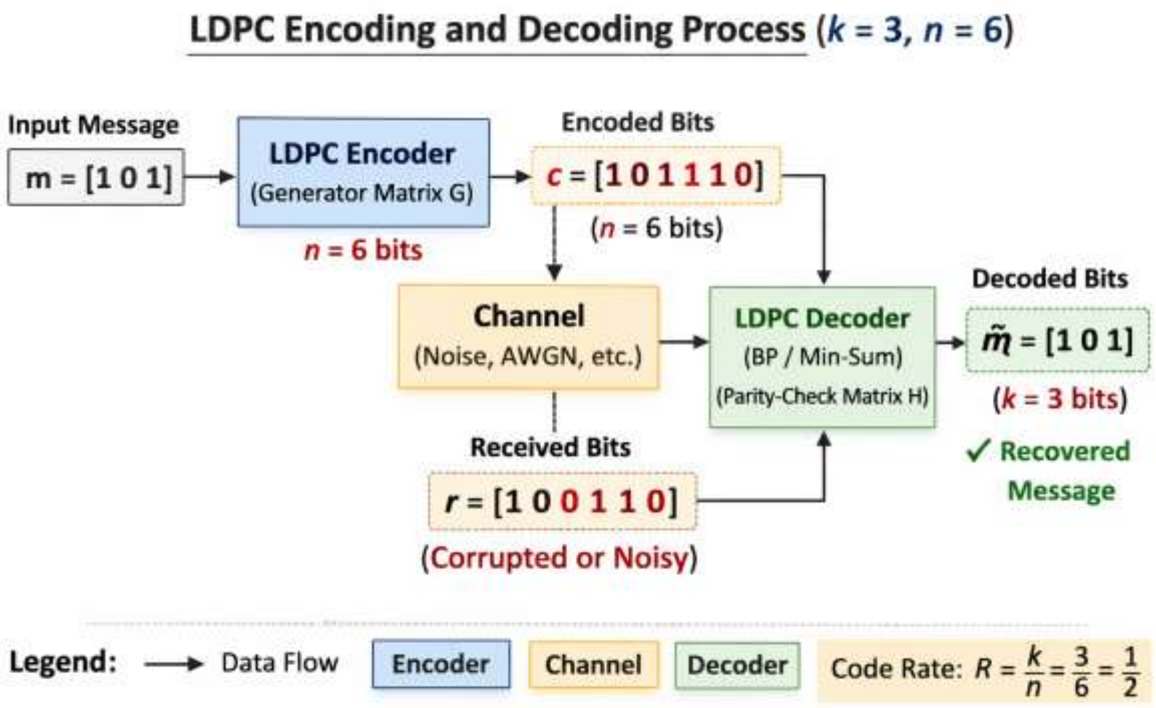
Cell Edge Scenario (remote users)

1. low SNR
2. weak signal
3. Highly sensitive to beam training errors

Remote users suffer due to low SNR + error propagation
Traditional beam training struggles in cell-edge scenarios

Coded beam training:

- Improves reliability
- Increases success rate
- Extends coverage
- Maintains efficiency



Transmission Rate < = Channel Capacity

R<=C (Shannon Channel Capacity)

C = Blog2(1 + SNR)

Message bits → 1011 (Tx) $\xrightarrow[\text{wireless channel}]{}$ 1111 (Rx) (noise)

[ ] → more 0's, less 1's

Error correction (Theoretical)

LDPC (Low Density Parity check) → adds extra bits → (sparse parity check matrix) → to fix errors using iterative decoding

dense matrix

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

sparse → spread out (0's, 1's)

$$H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

Ex →
$m = [1\ 0\ 1] \rightarrow k = 3$
Add 3 parity bits → total 6 bits
$c = [1\ 0\ 1\ p_1\ p_2\ p_3] \rightarrow n = 6$
$\boxed{R = 0.5}$ $R = \frac{k}{n} = \frac{3}{6}$ → code rate

$0.5 = \log_2(1+SNR)$
$2^{0.5} = 1 + SNR$
$1.414 = 1 + SNR$
$SNR = 0.414$
$\boxed{SNR_{min} = 0.414}$ → theoretical SNR

dB: $10\log_{10}(0.414) = -3.83\,dB$ → needed to transmit at rate 0.5

Shannon capacity
$C = B\log_2(1+SNR)$
$B = 1$
$R = 1.8$ $R \le C$ (reliable transmission)
$1.8 = \log_2(1+SNR)$
$2^{1.8} = 1 + SNR$
$\boxed{2^{1.8} = 3.48}$
$3.48 = 1 + SNR$
$SNR = 2.48$
$\boxed{SNR_{min} = 2.48}$

Transmission rate ≤ channel capacity

$p_1 = m_1 \oplus m_2 = 1 \oplus 0 = 1$
$p_2 = m_2 \oplus m_3 = 0 \oplus 1 = 1$
$p_3 = m_1 \oplus m_3 = 1 \oplus 1 = 0$

$c = [1\ 0\ 1\ 1\ 1\ 0]$

→ checks for errors multiple times and correct it.

(Practical)

1) $R = 0.5$ (fix code rate)
$R = \frac{k}{n}$

2) Generate data
Ex: 100000 random bits

3) Encode
$m = [1\ 0\ 1]$ $n = 6$ $c = [1\ 0\ 1\ 1\ 1\ 0]$

4) Pass through channel :- $y = x + n$
x → transmitted encoded bits
n → gaussian noise $n \sim N(0, \sigma^2)$

5) Decode :- Noisy y
LDPC → belief propagation → iteratively updates probability of each bit as 0 (or) 1
To correct the bits it uses → parity check (H matrix)
Output → $\hat{m}$ (1 0 1)

6) BER
Ex: 1000 errors → 100,000 bits → $BER = 0.01 = 10^{-2}$

SNR (dB)	BER
-4	10^{-2}
-3.6	10^{-3}
-3.2	10^{-4}
-3	10^{-5}

SNR ↑ → decoding becomes reliable

LDPC achieves $BER \le 10^{-5}$ at $-3\,dB$
Shannon SNR_{min} $R = 0.5$ = $-3.83\,dB$ } diff 0.83
(Near-Shannon performance)

Sum Rate

How fast data travels from base station (BS) to user device (UE) after beam training.

Coded training gives near-perfect speed (like testing all beams) but uses 98% less time.

Success Rate

Chance BS finds UE's exact best beam direction.

Coded fixes errors at weak signals, succeeds 90% like full tests but 14x quicker.

Multipath Effect

BS sends signal to UE, but it bounces off walls/buildings before arriving—creating "ghost" signals that mix with the direct path and confuse beam training.

Distance Effect

Farther UE = weaker BS signal.

Coded reaches 145m reliably (+25m beyond others) by correcting mistakes early.

Methodologies:

Adaptive Beam Training: System generates beam pattern based on signal feedback.

Non adaptive Beam Training: Uses pre computed beam throughout all stages.

Hierarchical Beam Training: Performs multi-stage beam search from wide beams to narrow beams for progressive refinement.

Traditional Hierarchical BT: A predefined multi-level beam refinement method where each stage depends on the previous beam selection.

Hard Beam Training: Selects the beam purely based on maximum received power without probabilistic metrics.

Soft Beam Training: Selects the beam using soft metrics like likelihood or LLR instead of raw power comparison.

Repetitive Beam Training: Transmits the same beam multiple times to improve reliability through power averaging.

Repetitive Code-Based Beam Training: Uses coded beam sequences and repeats them to enhance decoding reliability under noise.

Proposed CBT : Applies error-correcting codes to beam index detection for robust beam selection at low SNR.

Traditional Beam Training: Conventional beam search methods such as exhaustive or basic hierarchical scanning without coding enhancements

Exhaustive Beam Training: Sequentially tests every possible beam direction and selects the beam with the highest received power.

Considering Massive MIMO,

256 antennas
single user - LOS channel L = 1
Wavelength spacing - **λ/2**
Angle of arrival --> -90 to +90 deg (half plane in front of BS)
theta = 0 - user directly in front of BS
theta = 90 - user at far right edge of front half plane
theta = -90 - user at far left edge of front half plane

theta_DoA - azimuth angle of arrival of user signal to BS arrays

Error Correcting Codes:

BS sends beams; UE picks best via received power. Errors happen (noise/weak signal). ECC protects beam index like data bits.

Hamming Code: Adds parity bits to beam index bits to correct single-bit errors and prevent wrong beam selection.

Convolutional Code: Encodes beam index bits with memory and uses Viterbi decoding to reliably detect the correct beam under noise.

LDPC (Low-Density Parity-Check): Uses sparse parity-check matrices and iterative decoding to achieve very reliable beam detection at low SNR.

Polar Code: Uses channel polarization and successive cancellation decoding to reliably identify the correct beam with low complexity.

Turbo Code: Uses parallel convolutional encoders with iterative decoding to provide strong error correction for beam index detection.

Turbo Product Code (TPC): Combines two block codes in row-column structure to improve beam detection reliability through iterative decoding.

BCH Code:A powerful block code that corrects multiple bit errors in beam index bits using algebraic decoding.

Reed–Solomon (RS) Code:A symbol-based block code that corrects multiple symbol errors, making beam detection robust against burst noise.

System Model (Common to All Codes)

- Carrier frequency: **60 GHz**
- BS antenna array: **ULA**
- Antenna spacing: **λ/2**
- Users: **1**
- Beam index ↔ quantized angle
- Channel modeled using steering vectors

Exhaustive Beam Training

What the code does:

- Tests **all Nt beams**
- Selects beam with maximum received power

Why it exists:

- Provides **upper-bound accuracy**

Limitation:

- Extremely high training overhead

Traditional Hierarchical Beam Training

What the code does:

- Tree-based beam search
- Narrow beam region level-by-level
- Uses **hard decisions** at each stage

Limitation:

- One wrong decision propagates
- Performance degrades at low SNR

Repetitive Code-Based Beam Training

What the code does:

- Repeats each hierarchical beam transmission
- Averages received energy before decision

Why it helps:

- Reduces noise effect

Still limited because:

- No coding
- Still greedy and hard-decision based

Proposed Convolutional CBT

What the code does:

- Treats beam index bits as information bits
- Encodes them using **convolutional coding**
- Spreads information across hierarchy levels

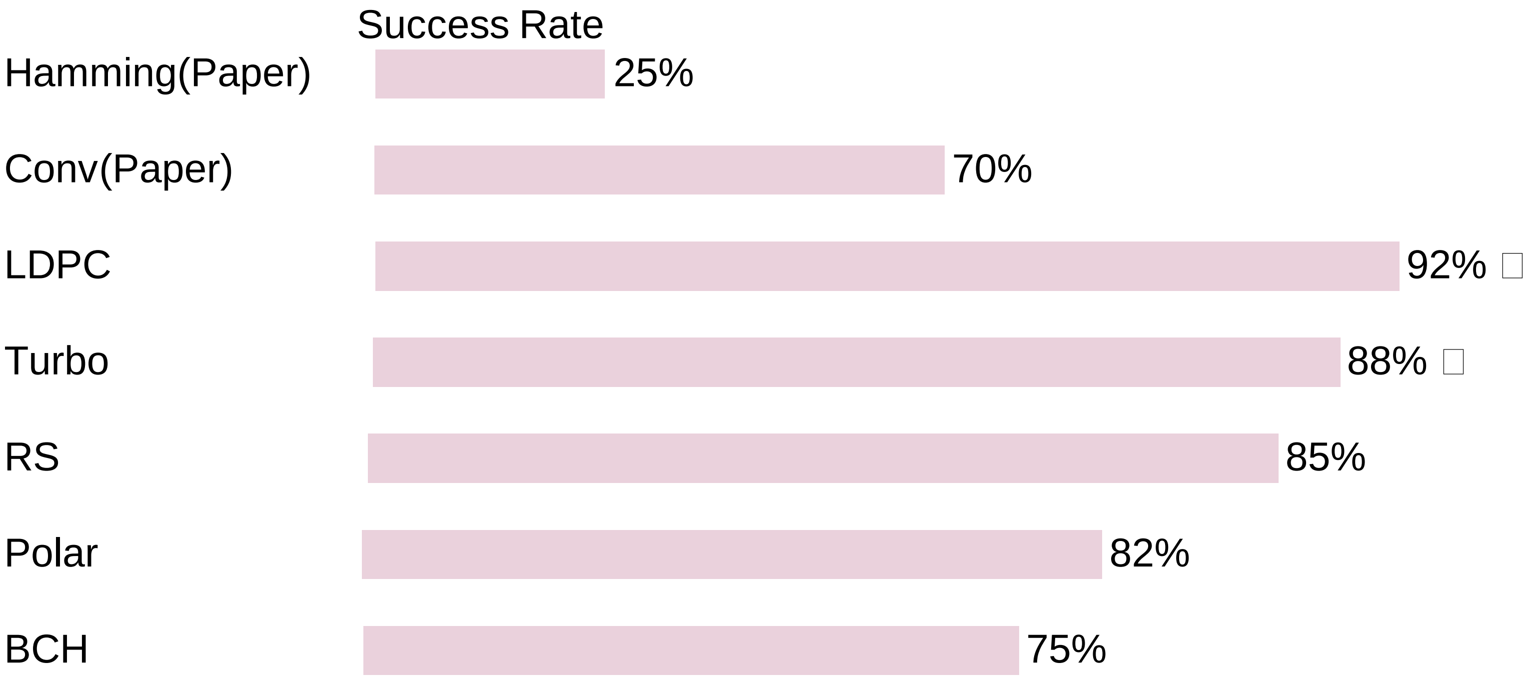
Receiver:

- Uses **soft metrics**
- Jointly decodes beam index

Performance Comparison

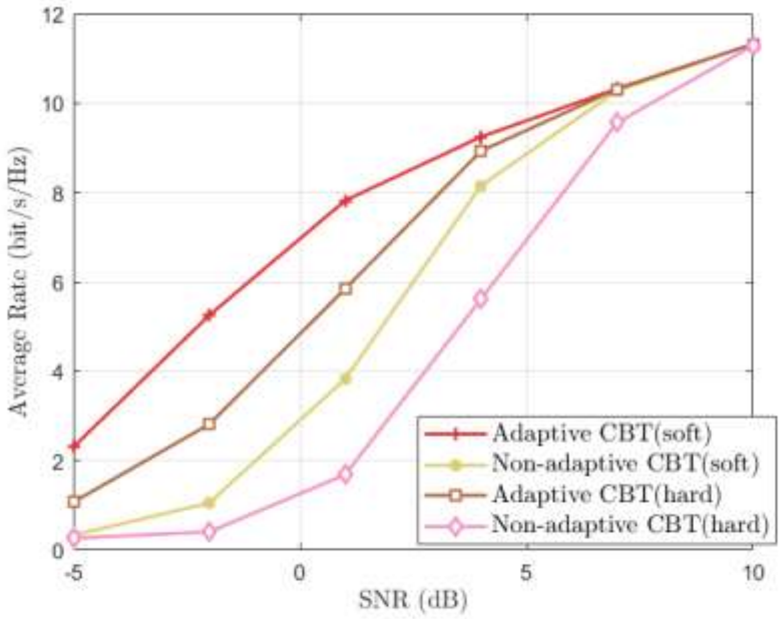
Code Type	Layers / Structure (for 1024 antennas)	SNR Gain (Approx.)	Best Use Case in Beam Training
LDPC	20 layers	+5 dB	Very large antenna arrays, strong coding gain
Polar	10 layers	+3 dB	Low-latency applications
Turbo	30 layers	+4 dB	Cell-edge users (low SNR)
Turbo Product Code (TPC)	126 component codes	+3 dB	Burst error scenarios
Reed–Solomon (RS)	32 symbols	+4 dB	Symbol-level error correction
BCH	20 parity bits	+2 dB	Low-complexity implementations

CELL-EDGE PERFORMANCE (-15dB SNR)



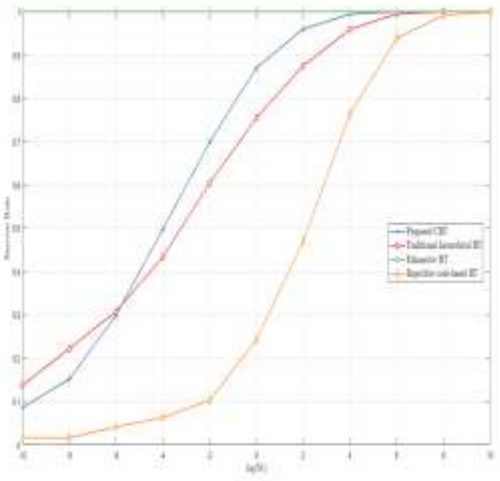
LDPC EXTENDS COVERAGE 3X!

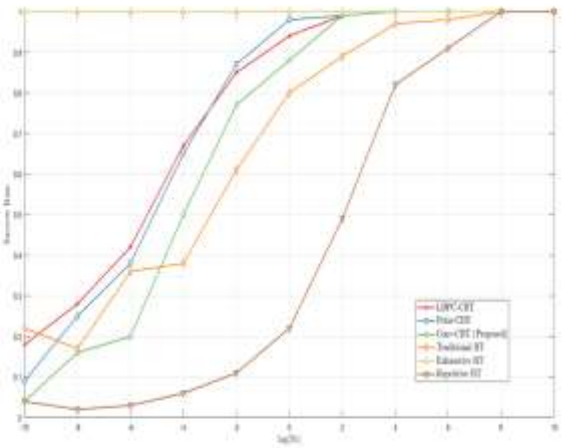
Fig 1 : main_sumrate



- Adaptive soft-decoded CBT achieves the highest spectral efficiency across all SNR levels.
- Soft decoding significantly improves noise robustness compared to hard decoding.
- Adaptive beam refinement enhances alignment accuracy over fixed hierarchical search.
- Performance gap is prominent at low SNR and converges at high SNR.
- Combining adaptive search with soft decoding yields the most reliable beam training strategy.

Fig 2 : main_successrate





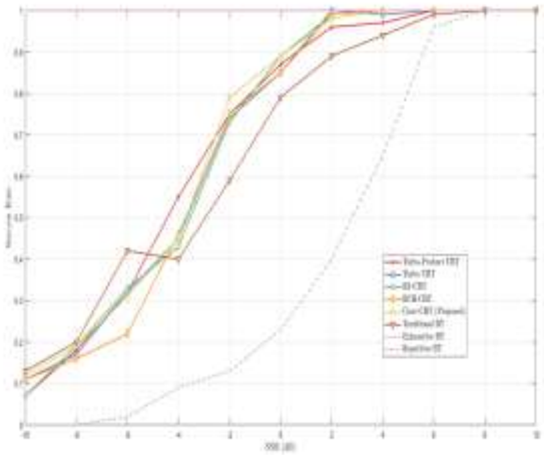
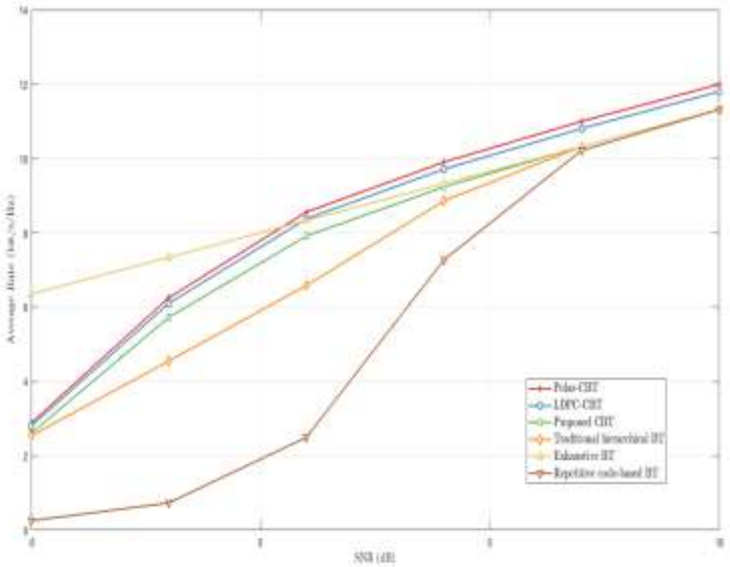


Fig 3 : main_sumrate_compare



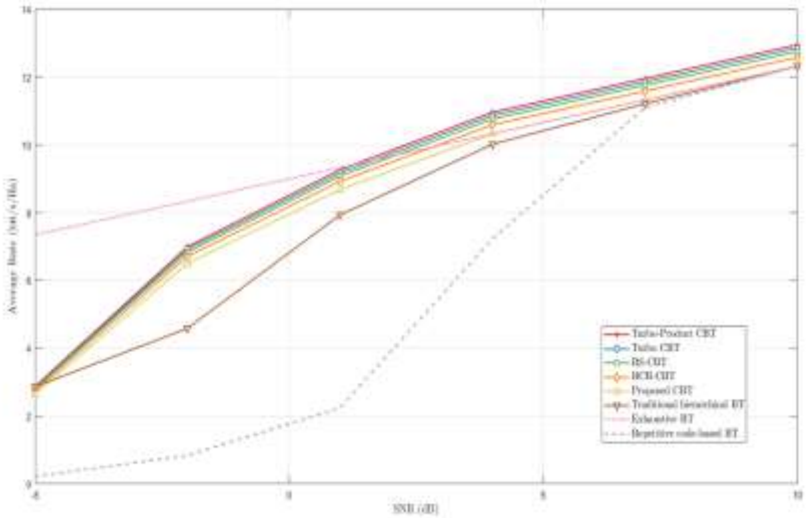
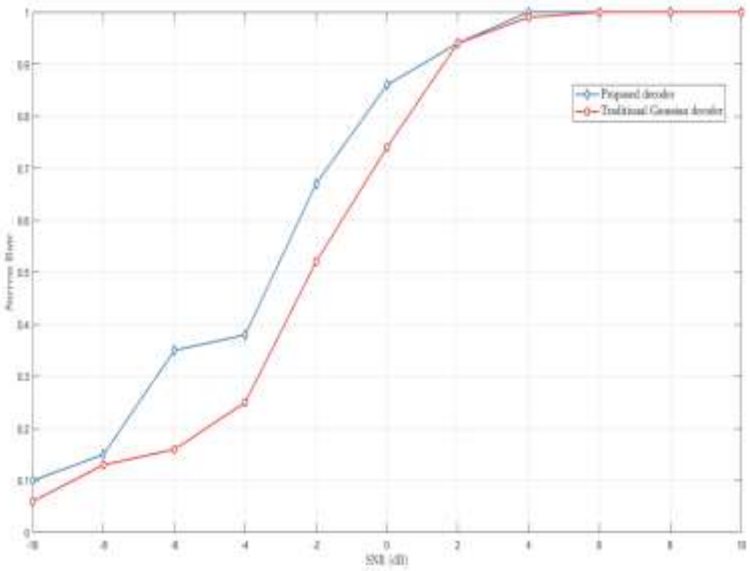


Fig 4 : main decoder



• Proposed convolution-coded hierarchical decoder significantly improves beam detection reliability at low SNR.

• Traditional Gaussian hierarchical decoding is more sensitive to noise.

• Performance gap is most visible in the low-SNR region.

• Both methods converge toward perfect detection at high SNR.

• Coding enhances robustness against noise-induced beam index errors.

Fig 5 : main_multipath

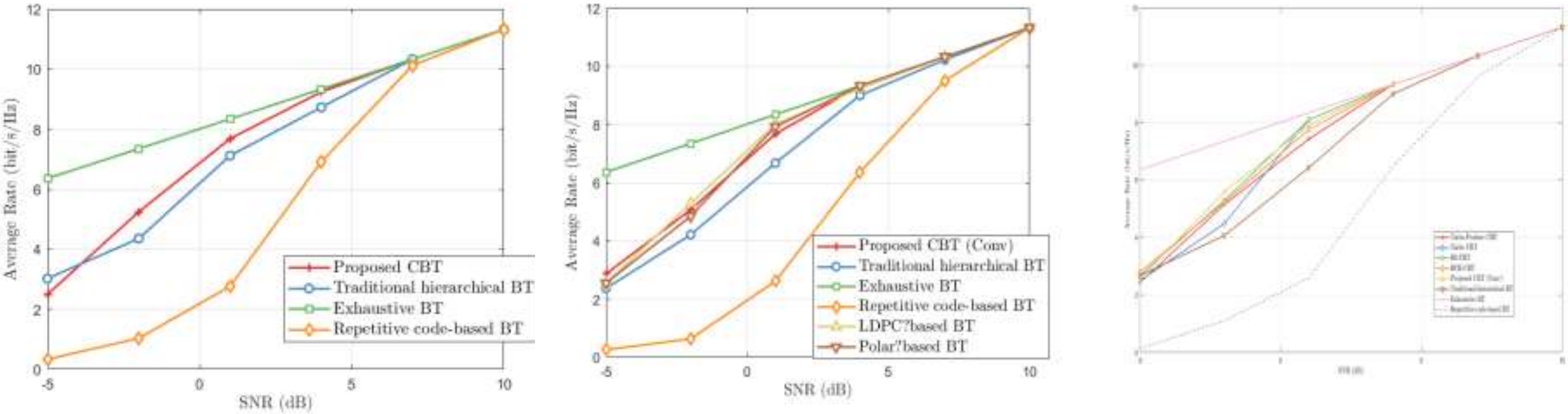
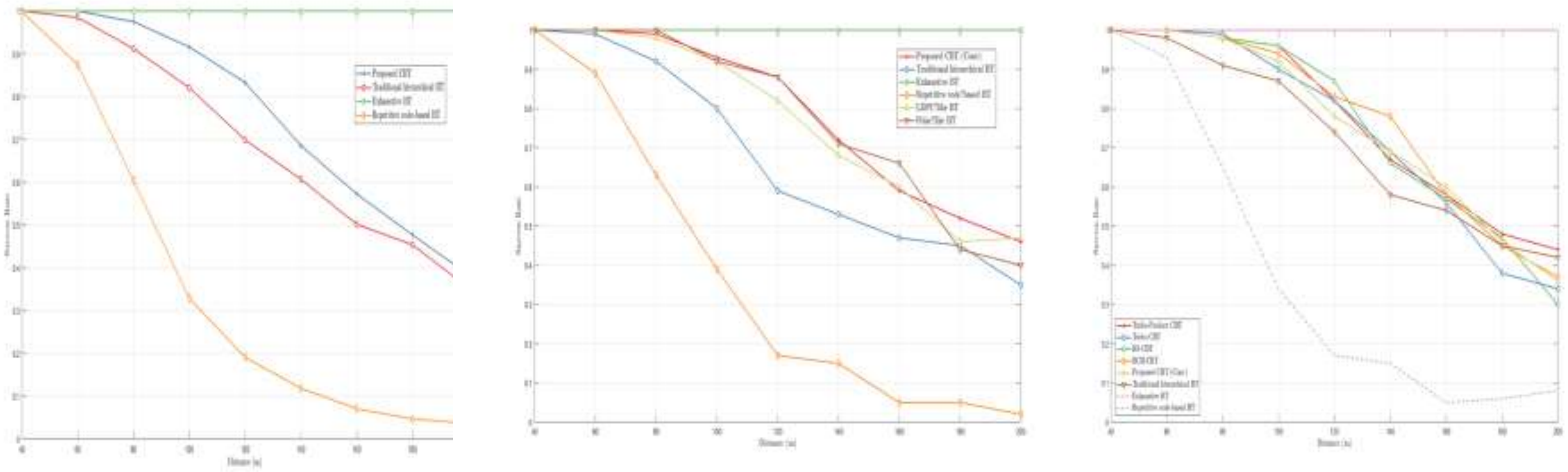


Fig 6 : main_distance



THANK YOU!